INTERACTIVE MAP WEB APPLICATION

Full Stack Web Development Project Report
(Unified Mentor Internship)

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1. Title Page

Project Title: Interactive Map
Domain: Full Stack Web Development (Frontend Mapping Application)
Technologies Used: HTML, CSS, JavaScript, Mapping Library (Leaflet / Google Maps API)
Difficulty Level: Hard
Internship Organization: Unified Mentor
Intern Name: *Vaibhav Thapliyal*
Duration: *Nov 2025 – Feb 2026*
Github: https://github.com/vai938/interactive_map

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2. Abstract

The Interactive Map Web Application is a browser-based mapping solution developed using HTML, CSS, and JavaScript, integrated with a third-party mapping library such as Leaflet or Google Maps API. The application displays multiple points of interest on an interactive map using customizable markers. Users can click on markers to view detailed information about each location, search for specific locations by name or category, and interact with the map through zooming and panning.

The project aims to enhance frontend development skills by introducing geospatial visualization, third-party API integration, and interactive UI design. The application is responsive and optimized for use on both desktop and mobile devices, providing an intuitive user experience for exploring mapped locations.

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3. Introduction

Interactive maps are widely used in modern web applications for navigation, location discovery, and spatial data visualization. From ride-hailing platforms to tourism websites, map-based interfaces have become a standard UI pattern.

This project was developed as part of the Unified Mentor internship to provide hands-on experience with integrating mapping libraries into web applications and managing interactive elements such as markers, popups, and search functionalities. The project demonstrates how geospatial data can be presented in a visually appealing and user-friendly manner using frontend technologies.

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4. Problem Statement

Users often require intuitive interfaces to explore locations, points of interest, or categorized places within a geographical context. Static maps lack interactivity and provide limited user engagement.

Problem:
To design and implement an interactive, responsive web-based map that displays multiple locations with clickable markers, provides search functionality, and presents contextual information for each point of interest.

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5. Objectives

The main objectives of the Interactive Map project are:

- To integrate a mapping library (e.g., Leaflet or Google Maps) into a web application.
- To display multiple markers representing different locations.
- To implement interactive popups for location details.
- To provide search functionality for locating specific places or categories.
- To design a responsive and visually appealing user interface.

- To improve understanding of third-party API usage in frontend development.

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6. Scope of the Project

In Scope:

- Interactive map display
- Minimum of five markers for points of interest
- Clickable markers with information popups
- Search bar for location lookup
- Responsive UI for mobile and desktop
- Custom styling using CSS

Out of Scope (Future Enhancements):

- Real-time location tracking
- Backend integration for dynamic location data
- User accounts and saved locations
- Route planning and navigation

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7. Literature Review (Brief)

Modern web mapping applications rely on JavaScript libraries such as Leaflet, Mapbox GL JS, and Google Maps API. These tools provide abstractions for rendering tiles, handling zoom and pan interactions, and managing marker layers. Interactive mapping improves user engagement and enables spatial data exploration in numerous application domains including tourism, logistics, and urban planning.

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8. System Architecture

High-Level Architecture:

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User Interaction
        ↓
HTML/CSS UI (Map Container + Search Bar)
        ↓
JavaScript Controller
        ↓
Mapping Library (Leaflet / Google Maps API)
        ↓
Map Tiles + Marker Data
        ↑
Popup Rendering & UI Updates
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Core Components:

- UI Layer (Map container, controls, search bar)
- Map Rendering Engine (Leaflet / Google Maps)
- Marker Management Module
- Search & Filter Module

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9. Technology Stack

Layer	Technology Used
Frontend	HTML5, CSS3

Scripting	JavaScript (ES6)
Mapping	Leaflet / Google Maps API
Tools	VS Code, Browser

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10. Functional Requirements

- The system shall display an interactive map.
- The system shall render at least five markers.
- The system shall display popups with location details on marker click.
- The system shall allow users to search locations by name or category.
- The system shall center the map on searched locations.
- The system shall support zooming and panning interactions.

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11. Non-Functional Requirements

- **Usability:** Intuitive and minimal UI controls.
- **Responsiveness:** Optimized for mobile and desktop screens.
- **Performance:** Smooth map rendering and marker interactions.
- **Reliability:** Accurate marker placement and search results.
- **Maintainability:** Modular JavaScript code for easy enhancements.

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12. Implementation Details

12.1 HTML Structure

HTML defines:

- A container element for the map

- A search bar for user queries
- A panel or popup container for location details

12.2 CSS Styling

CSS is used to:

- Style the map container
- Customize marker popups
- Design responsive layout
- Enhance UI aesthetics

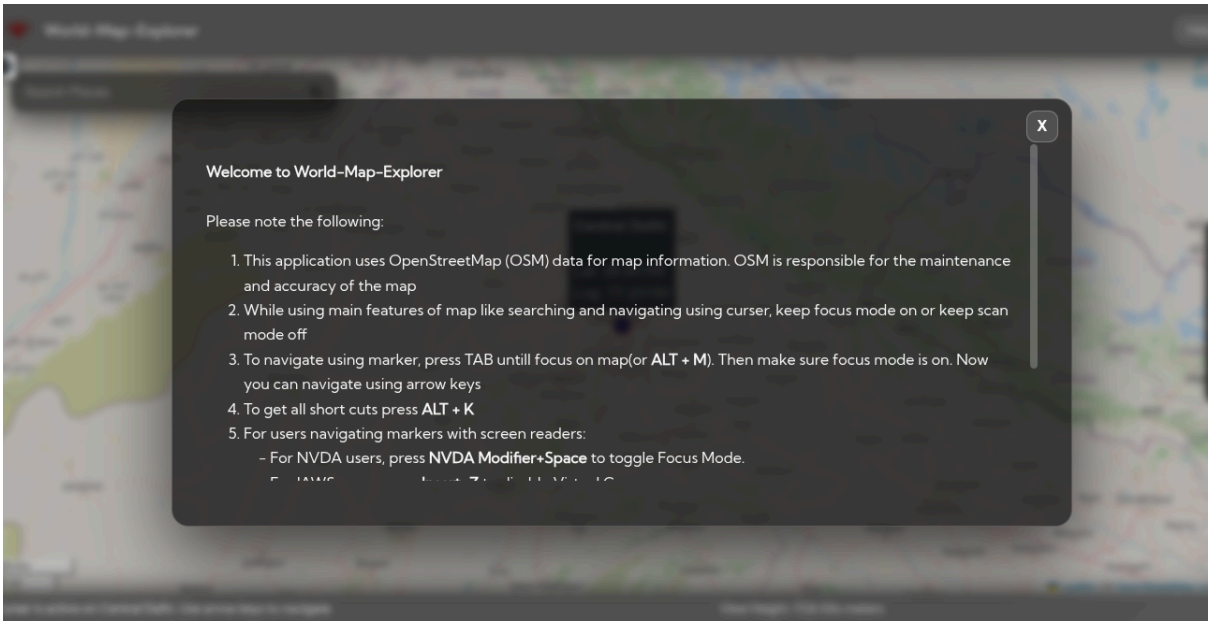
12.3 JavaScript Logic

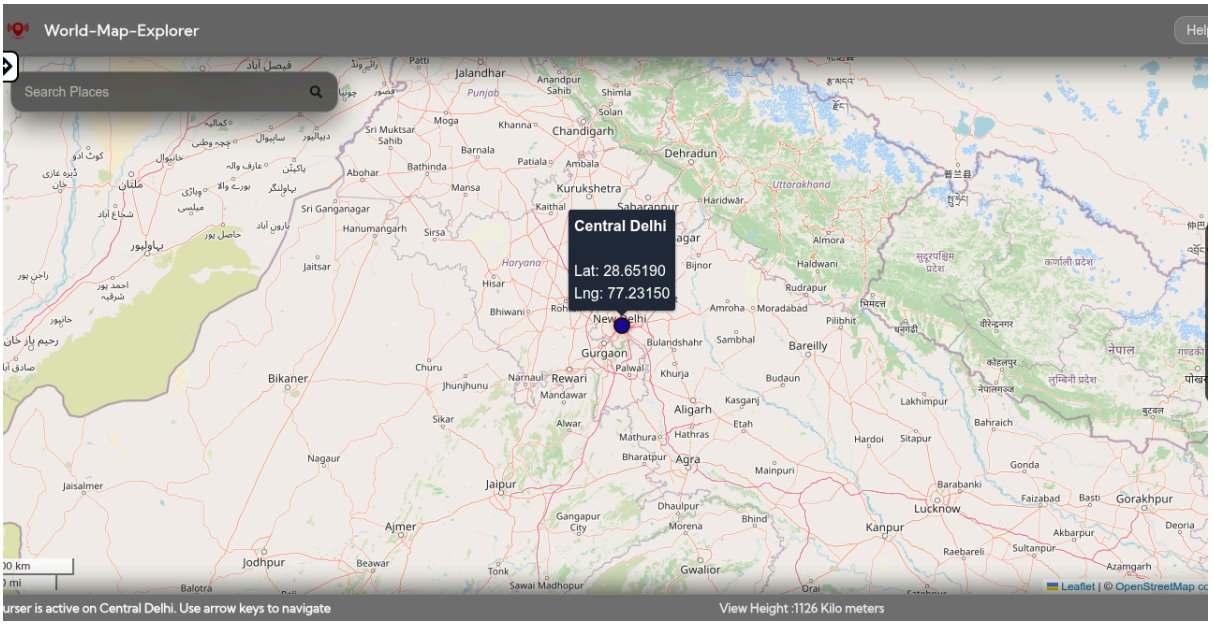
JavaScript manages:

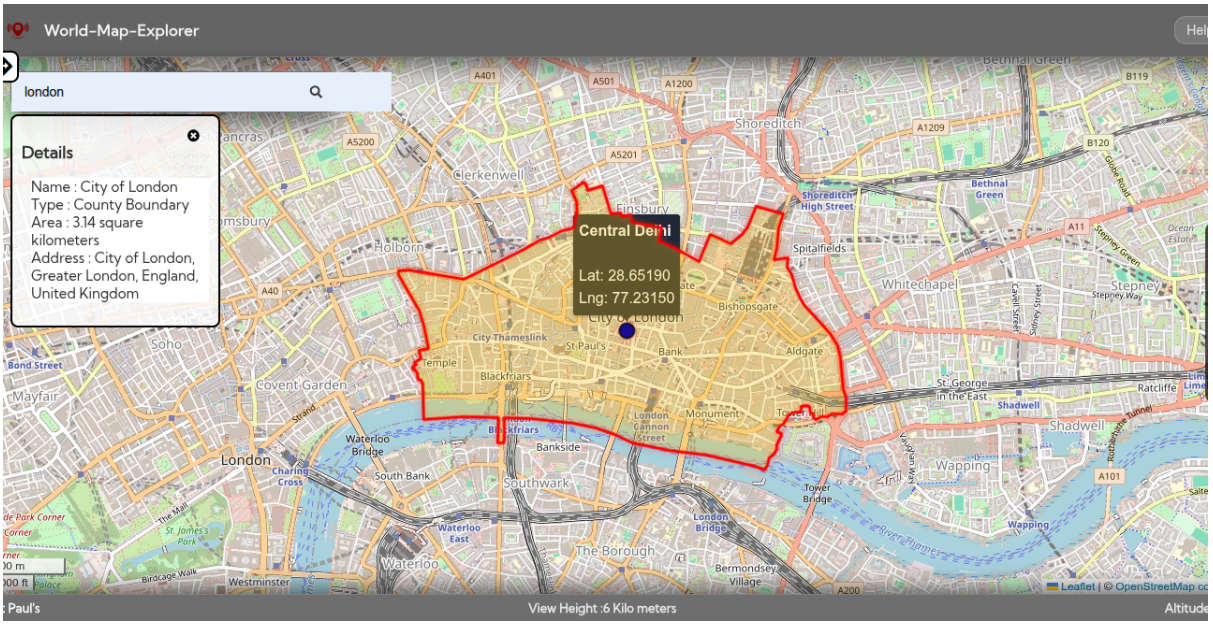
- Initializing the map using the chosen mapping library
- Adding and customizing markers
- Binding popup information to markers
- Implementing search and centering logic
- Handling responsive resizing events

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13. Output Screens (Placeholders)







- Main Map View
- Marker Popup Display
- Search Results Centered on Map
- Mobile View of the Map

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14. Testing

Testing activities included:

- Verifying marker placement accuracy

- Testing marker click interactions
- Validating search functionality
- Checking responsiveness on different devices
- Cross-browser compatibility testing

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15. Challenges Faced

- Integrating and configuring the mapping library
- Managing API keys (if required)
- Customizing marker icons and popups
- Ensuring smooth performance on mobile devices
- Implementing accurate search and filtering logic

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16. Results and Outcomes

The Interactive Map application successfully displays multiple locations with interactive markers and informative popups. Users can search for locations and explore points of interest in a responsive, visually appealing interface. The project strengthened practical understanding of:

- Third-party API integration
- Geospatial UI design
- Interactive frontend development
- Event-driven programming

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17. Future Enhancements

- Add geolocation to show user’s current position
- Implement category-based filters
- Add route navigation and directions
- Integrate backend for dynamic location data
- Add animations and transitions for enhanced UX

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18. Conclusion

The Interactive Map project demonstrates the effective integration of mapping libraries with frontend web technologies to create interactive, data-rich web applications. The project fulfills all defined requirements and provides a scalable foundation for building more advanced geospatial web solutions. It contributed significantly to practical learning outcomes during the Unified Mentor internship.
